1. **(5 points)** Starting with the Euler's relationship: $e^{j\theta} = \cos\theta + j\sin\theta$ show that $\cos 2x = \cos^2 x - \sin^2 x$

2. **(5 Points)** Compute the exponential Fourier series of an impulse train $x(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_0)$. Now use the computed Fourier series to find the Fourier transform of the impulse train. What can you conclude from the result?

3. **(10 Points)** Starting with the original sine/cosine Fourier representation,

$$x(t) = a_0 + \sum_{n=0}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

derive the exponential form of the Fourier series

$$x(t) = \sum_{n=-\infty}^{\infty} x_n e^{jn\omega_0 t}$$

with $x_n = \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} x(t) e^{-jn\omega_0 t} dt$. **For full credit you need to show the derivation of the expression for x_n too. This means, you need to show that** $x_n = \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} x(t) e^{-jn\omega_0 t} dt$

4. **(5 Points)** From the above, derive the Parseval's theorem for the exponential form of the Fourier series.

5. **(5 Points)** Show that the Fourier transform of $\int_{-\infty}^{t} x(t)dt = \frac{X(\omega)}{j\omega} + \pi X(0)\delta(\omega)$. How do you compute $X(0)$?

6. **(5 Points)** Show that the Fourier transform for an even signal $x(t)$ is given by

$$X(\omega) = 2\int_{0}^{\infty} x(t)\cos(\omega t)dt$$

7. **(5 Points)** Compute the Fourier transform of this signal: $x(t) = e^{-|t|}\cos(\omega_0 t)$. Use the properties of Fourier transform.

8. (10 Points) Compute the Fourier transform of the following signal

$$x(t) = e^{-at}\cos(\omega_0 t)u(t)$$